ENERGY STORAGE · PRODUCT & ENGINEERING

Modular Energy Storage

One architecture from portable power to home energy — P1 Go, P2 Home, Energy Module, Pod and Energy OS. Tesla-grade experience integration. BYD-grade module repetition.

Document ID: PF-MODENERGY-001

Version: 1.0 · 30 August 2026

Status: Product design foundation + engineering design basis (not issued-for-construction)

Contents: Part I Product Design · Part II Engineering Design (Tesla/BYD architecture motivation)

> **Design principle:** Energy storage should be a positive experience — simple to use, easy to move, easy to expand, and ready to connect to solar, home and electric mobility.
>
> **PLUG → CARRY → ROLL → STACK → INSTALL → DRIVE**

PART I · PRODUCT DESIGN

One Architecture. From Portable Power to Home Energy.

Document: Modular Energy Platform — Product Design Page **Version:** 1.0 **Status:** Product Design Foundation (companion to PF-MODENERGY-002) **Placement:** Product design companion to PF-MODENERGY-002 (see Canon and product docs)

> **Canonical engineering architecture: Modular Energy Architecture v2** (PF-MODENERGY-002 v2.0) — derived math, ERNs, Kenya design basis, 24 figures, 8 renders, prototype roadmap. **PDF:** precifarm-modular-energy-platform-v2.pdf · regenerate: `npm run pdf:modular-energy-v2` from `website/`.

Related: Architecture v2 · Industrial Design Team Brief · Product Overview · Engineering Design v1 · EV Charging index · Product doc index · Canon

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Energy storage should be a **positive experience** — simple to use, easy to move, easy to expand, and ready to connect to solar, home and electric mobility.

The physical experience evolves through:

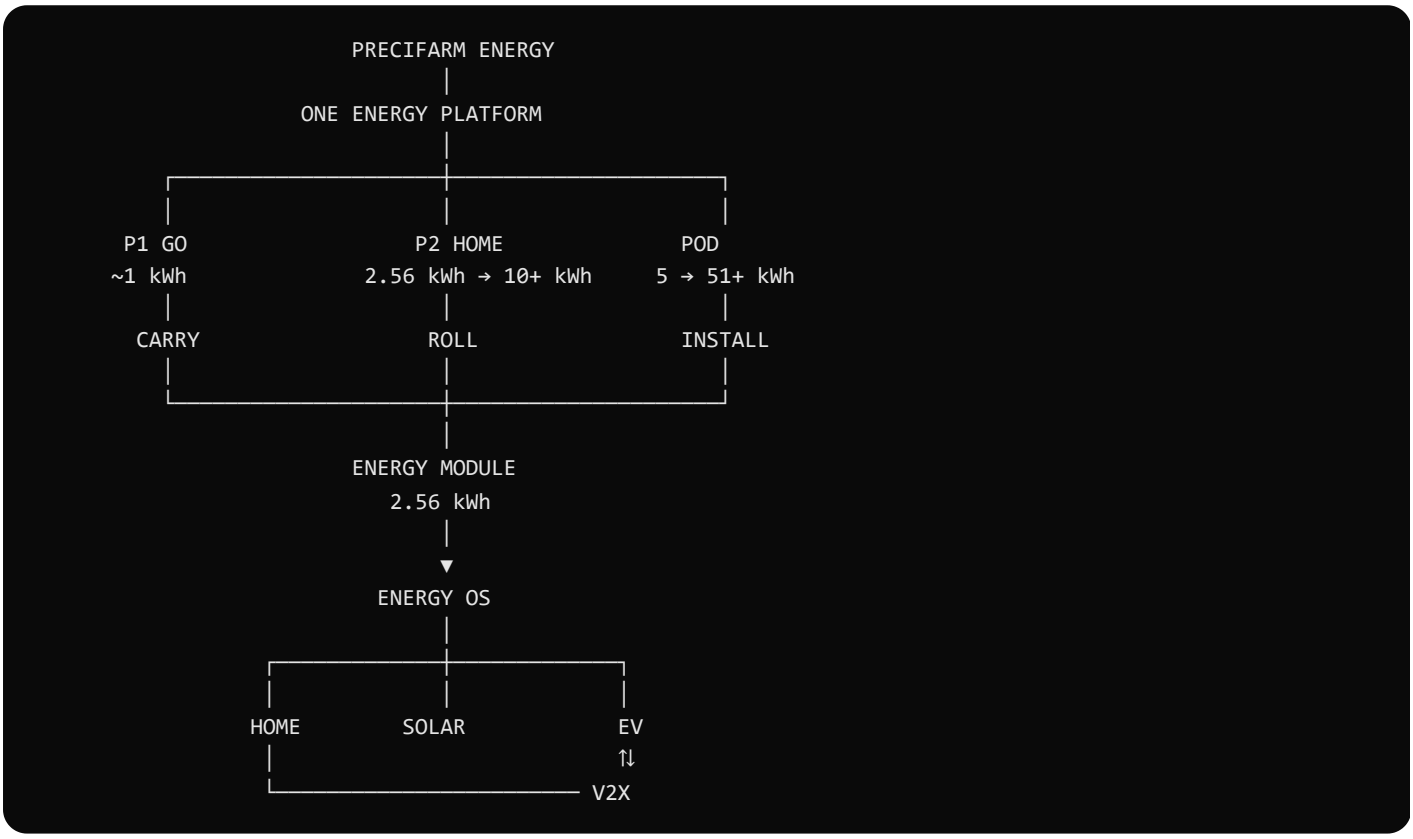
PLUG → CARRY → ROLL → STACK → INSTALL → DRIVE

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01. The modular energy principle

Precifarm does not design separate batteries for separate products.

Precifarm designs **one modular energy platform** that can take different physical forms as the customer's energy needs grow.



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02. One platform — many scales

The Precifarm portfolio should not be treated as a collection of unrelated batteries. It should be understood as a progression:

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Product	Target capacity	Physical experience	Primary role
P1 Go	~1 kWh	Carry	Portable energy
P2 Home	2.56–10+ kWh	Roll	Modular home energy
Energy Module	2.56 kWh	Stack	Common energy building block
Precifarm Pod	5–51+ kWh	Install	Scalable energy infrastructure
EV	Vehicle-dependent	Drive	Mobile energy resource

The customer can enter at a small scale and progressively increase energy capacity and functionality.

Start small. Add energy. Grow into a complete energy system.

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03. The Precifarm Energy Module

The Energy Module is the foundation of the modular architecture.

Target module: 2.56 kWh $LiFePO_4$ · 51.2 V × 50 Ah



The module should be designed as a standardized, serviceable energy building block.

Potential characteristics:

- $LiFePO_4$ chemistry
- Integrated BMS
- Cell monitoring
- Temperature monitoring
- SOC/SOH estimation
- CAN communication
- Mechanical locking

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- Module identification
- Service diagnostics
- Integrated handling features

The customer should not need to understand the underlying electrical architecture.

The customer should simply see:

ADD ENERGY

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04. One module — many systems

Where technically and economically feasible, the same Energy Module architecture should support multiple Precifarm products.



Example capacity progression

Configuration	Capacity	Experience
P2	2.56 kWh	Start
P2 + 1 Module	5.12 kWh	Expand
P2 + 2 Modules	7.68 kWh	Grow
P2 + 3 Modules	10.24 kWh	Scale
Pod Base	5.12 kWh	Install
Pod Array	10–51+ kWh	Infrastructure

Final module count, voltage architecture, inverter topology and expansion limits remain engineering and certification decisions.

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05. P1 Go — carry

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P1 Go is the entry point into the Precifarm energy ecosystem. It should be compact, premium and genuinely portable.



Core characteristics

- Integrated carry handle
- Compact footprint
- $LiFePO_4$ battery
- AC output · USB-C · USB-A · DC output
- Portable solar input · grid charging
- UPS/pass-through capability
- Wi-Fi · Bluetooth · energy monitoring
- Precifarm Energy OS

Customer experience: UNBOX → CHARGE → USE

P1 Go should feel like a premium energy appliance, not a battery.

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06. P2 Home — roll

Expandable Mobile Home Energy · Target base capacity: 2.56 kWh

P2 Home is the hero product and the bridge between portable power and installed energy infrastructure.



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P2 should combine:

- Integrated wheels · pull handle · secondary lifting handles
- Low center of gravity · stable base · protected ventilation
- Front-facing user interface
- Modular battery architecture
- Solar integration · grid charging · UPS capability
- Remote monitoring · Energy OS
- M-Pesa / PAYGO capability
- Future EV integration

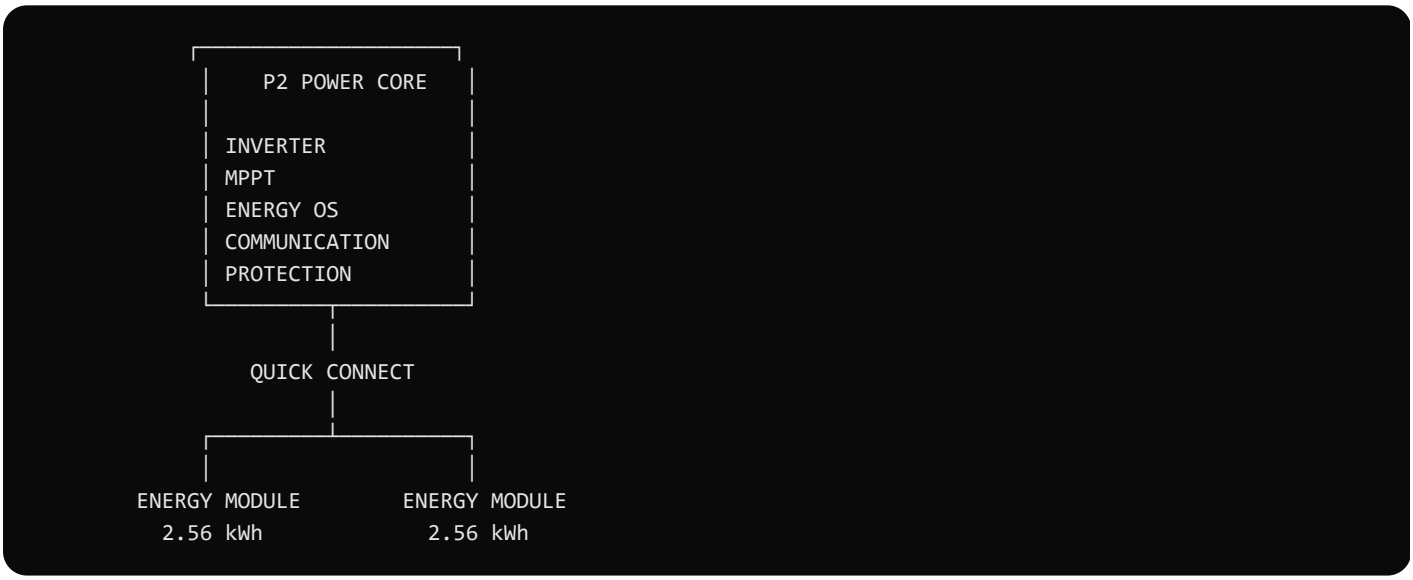
Customer experience: PLUG → USE → ADD ENERGY

P2 should be positioned as: **Your mobile home energy system.**

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07. Power Core + Energy Module

P2 should separate power electronics from expandable energy capacity.



Power Core — systems required to convert, control and manage energy:

Inverter · MPPT · AC charging · protection · communications · Energy OS gateway · system controller · thermal management

Energy Module — energy storage capacity:

$LiFePO_4$ cells · BMS · temperature sensing · SOC/SOH monitoring · CAN communication · protection · mechanical and electrical interfaces

This separation improves modularity, serviceability, upgradeability, manufacturing flexibility and product-family scalability.

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Precifarm modularity is not only about adding battery capacity. It is also about reducing installation complexity.

Product	Journey	Experience
P1 Go	UNBOX → CHARGE → USE	Plug-and-play portable energy
P2 Home	PLUG → USE → EXPAND	Plug-and-grow home energy
Pod	SURVEY → INSTALL → COMMISSION → EXPAND	Professionally engineered energy infrastructure

```
P1          P2              POD
PLUG & PLAY → PLUG & GROW → DESIGN & SCALE
```

The product experience should become progressively more powerful without becoming unnecessarily complicated.

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09. Solar integration

Solar should progressively increase in importance across the product family.

Stage	Flow
P1 Go	Portable solar charging — `SOLAR → P1`
P2 Home	Home solar + storage — `SOLAR → P2 → HOME`
Pod	Hybrid solar + storage + grid + EV infrastructure

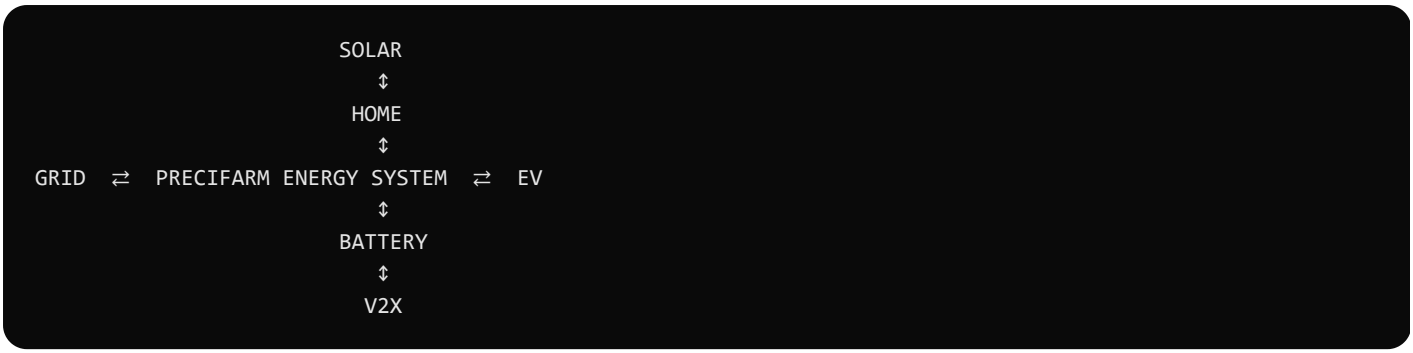


The customer progressively moves from portable energy into a complete energy ecosystem.

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10. Energy can move

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Future bidirectional applications may include:

Mode	Description
V2H	Vehicle → Home
V2L	Vehicle → Load
V2V	Vehicle → Vehicle
V2G	Vehicle → Grid

Availability of each function will depend on compatible vehicles, bidirectional hardware, utility requirements, standards and regulation.

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11. Precifarm Pod — install

Modular Energy Infrastructure · Target base capacity: 5.12 kWh

The Pod takes the same modular philosophy and converts it into a professionally installed energy system.



Potential applications: homes · SMEs · commercial buildings · solar-storage systems · EV charging · backup power · energy management · microgrids

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12. Energy capacity + power capacity

Precifarm should deliberately separate **energy capacity** from **power capacity**. This allows the system to scale according to different customer requirements.

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BATTERY CAPACITY          POWER CAPACITY
5 → 10 → 20 → 30 → 50+ kWh    +    4 → 6 → 10 → 15 → 30+ kW
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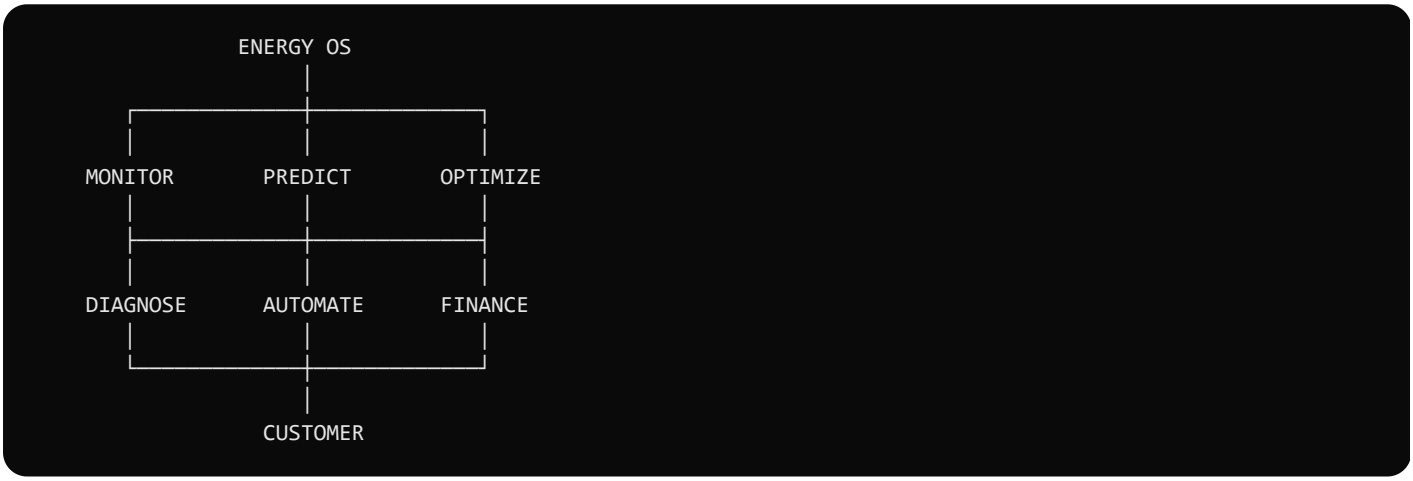
Example requirements: 10 kWh / 10 kW **or** 30 kWh / 10 kW

The final architecture should allow appropriate combinations of storage capacity and power electronics.

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13. Energy OS

Every Precifarm energy product should connect to the same digital layer.



Core capabilities

- Real-time monitoring · runtime prediction · solar forecasting · load optimization
- Battery protection · remote diagnostics · automated energy management
- PAYGO management · M-Pesa integration · service management
- Capacity recommendations · EV energy management · V2X coordination

The customer sees:

- How much energy do I have?
- How long will it last?
- Where is my energy coming from?
- Where should it go?

The complexity stays underneath.

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Precifarm should not make customers think about technical infrastructure unless they want to. The system should translate technical information into useful decisions.

Instead of	Show
Battery: 63%	6h 18m remaining
Solar input: 1.2 kW	Solar is powering your home and charging your battery.
Expansion battery detected	+2.56 kWh available
V2H active	Your vehicle is helping power your home.

Goal: Make energy understandable, predictable and useful.

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15. Energy should feel positive

Precifarm should differentiate itself from traditional battery and BESS companies through the experience it creates.

Traditional energy-storage language focuses on: battery capacity · inverter efficiency · peak shaving · backup duration · energy arbitrage · battery chemistry

Precifarm should translate these into customer outcomes:

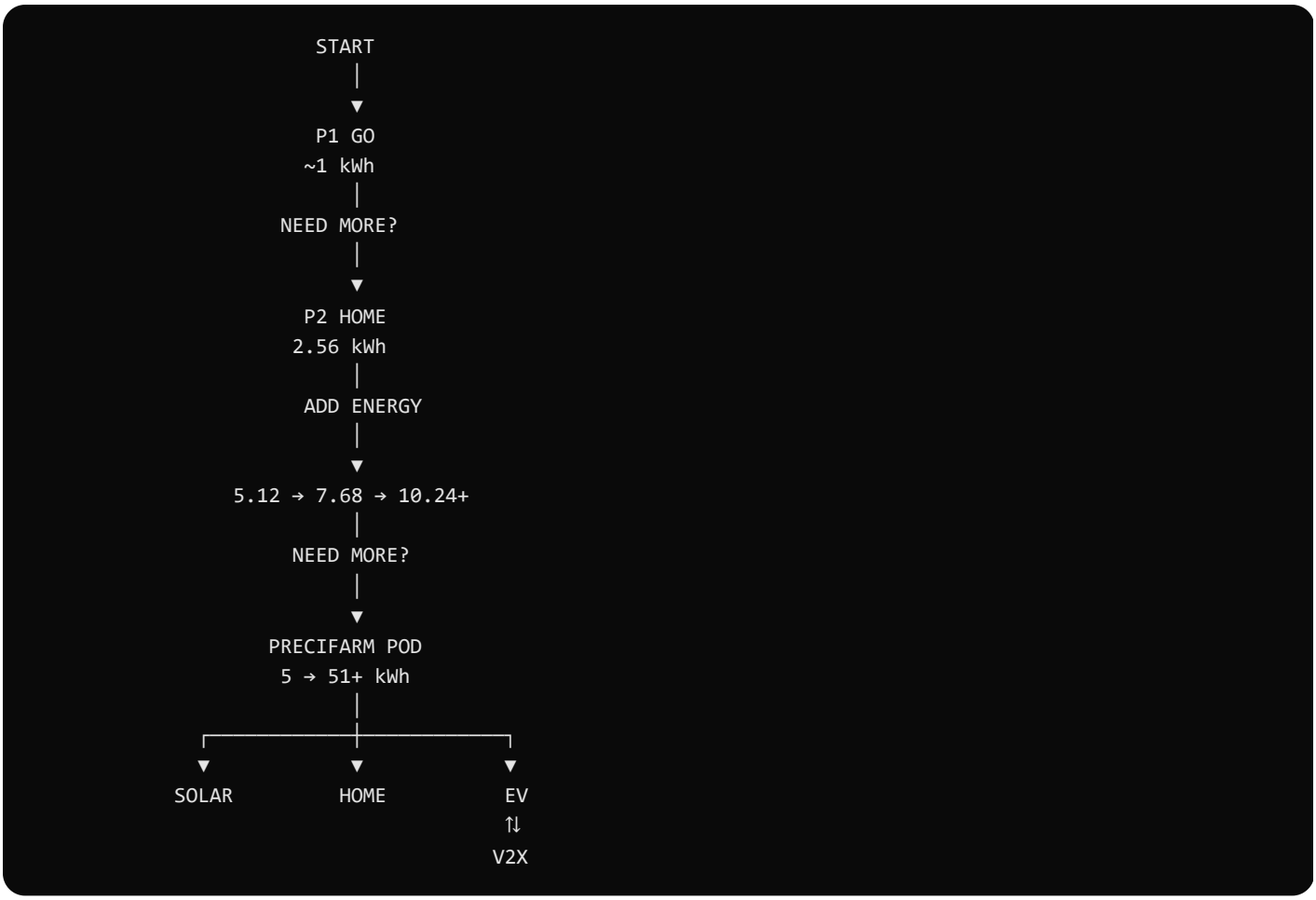
- My home stays powered.
- My solar charges my battery.
- I can add more energy when I need it.
- I can see where my energy is going.
- My battery protects my home during an outage.
- My vehicle can eventually help power my home.

The customer should experience capability, not complexity.

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16. The customer journey

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Commercial principle: Buy what you need today. Add capability when your life changes.

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17. Product family design language

All Precifarm energy products should immediately look like members of the same family.

Shared design DNA

Common geometry · rounded industrial forms · consistent proportions · common materials · consistent typography · consistent iconography · consistent status indicators · consistent port markings · consistent connector philosophy · consistent Energy OS interface · common battery terminology · common service architecture

Product-specific characteristics

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Product	Primary physical characteristic
P1 Go	Compact + carryable
P2 Home	Wheeled + mobile
Energy Module	Stackable + serviceable
Pod	Fixed + scalable
EV Interface	Bidirectional + connected

The products should share DNA without becoming identical.

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18. Physical experience

The product family should communicate its progression visually.



The complete physical progression is:

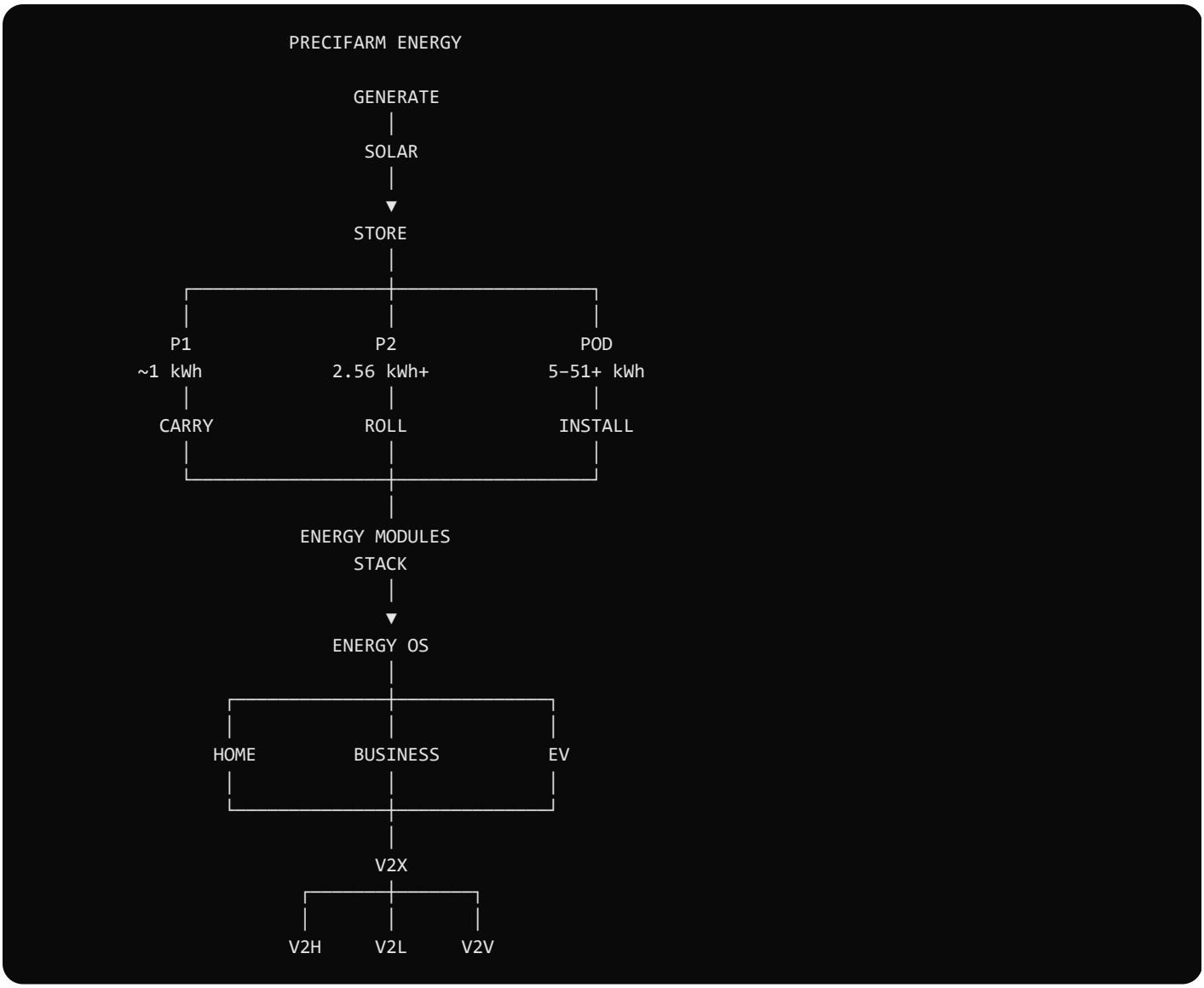
CARRY → ROLL → STACK → INSTALL → DRIVE

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19. The Precifarm energy network

The ultimate objective is not to create a collection of batteries. It is to create an energy network in which energy can be generated, stored, managed and moved.

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20. Design moat

Precifarm does not need to manufacture every component.

Buy: $LiFePO_4$ cells · power electronics · inverter components · standard communication components · displays · connectors · protection components

Customize: Enclosures · mechanical architecture · Energy Modules · harnesses · thermal architecture · user interface · local installation architecture

Own: Energy OS · energy data · runtime algorithms · energy optimization · diagnostics · PAYGO platform · M-Pesa integration · customer account · service network · modular product architecture · customer energy experience

The objective is to build defensibility around the system and experience, not necessarily around commodity components.

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Precifarm makes energy modular.

Start with portable power. Expand into home energy. Add solar. Add storage. Scale into commercial energy infrastructure. Connect an EV. Eventually allow energy to move between vehicles, homes, businesses and the grid.

Precifarm makes energy modular, mobile and intelligent.

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22. The master design principle

ONE PLATFORM. MANY SCALES.

Energy storage should be a positive experience.

It should be easy to acquire. Easy to use. Easy to move. Easy to expand. Easy to understand.

It should work with solar. It should support the home. It should connect to electric mobility.

And, where technically and commercially possible, energy should be able to move wherever it creates the most value.

```
PRECIFARM
PLUG → CARRY → ROLL → STACK → INSTALL → DRIVE
P1 GO → P2 HOME → ENERGY MODULE → POD → EV
ONE ENERGY OS. ONE CUSTOMER ACCOUNT. ONE MODULAR ENERGY PLATFORM.
```

Engineering references: ***Modular Energy Architecture v2*** *(primary) · Modular Energy Engineering Design v1 · EV Charging index. Product framing: Product Overview · Canon.*

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PART II · ENGINEERING DESIGN

Document ID: PF-ENG-MODENERGY-001 **Version:** 1.0 · 30 August 2026 **Owner:** Precifarm Engineering **Status:** Superseded for engineering targets and calculations — retained as historical basis **Superseded by: Modular Energy Architecture v2** (PF-MODENERGY-002 v2.0) **Product design companion:** Modular Energy Platform **Canon:** Consumer/home energy is a **gated expansion surface**; route Reserved Route Charging remains the beachhead. See CANON.md.

> **Use v2 for:** module math, ERNs, dual architecture (LV-48 / HV-S), Kenya design basis, PAYGO policy, test matrix, risk register, and PDF export. This v1.0 file remains for early Tesla/BYD motivation and block-diagram history.

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1. Purpose

This document translates the Modular Energy Platform product vision into an **engineering architecture** Precifarm can prototype, certify and manufacture through partners.

It answers:

- What is the common **Energy Module** electrical and mechanical interface?
- How do **P1 Go**, **P2 Home** and **Pod** reuse that module without forking the BMS/CAN stack?
- What do we **buy**, **customize** and **own** at the system level?
- What safety, grid and Kenya regulatory envelopes apply?
- What do **Tesla** and **BYD** teach — and what Precifarm should **not** copy?

1.1 In scope

- Reference architecture, block diagrams, interface definitions
- Target module specification (2.56 kWh $LiFePO_4$ class)
- Power Core vs Energy Module separation (P2 / Pod)
- Communications, Energy OS boundary, OTA and diagnostics
- Thermal, protection, EMC and service architecture
- Kenya installation classes (home plug-in vs fixed Pod)
- Phased prototype → pilot → certification roadmap

1.2 Out of scope

- Issued-for-construction wiring diagrams and protection coordination studies
- Final cell supplier selection and UN38.3 / IEC 62619 test reports
- Vehicle-side V2X hardware (referenced as future interface only)
- Corridor DC hub design (see Solar Chargers & Stations)

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Precifarm is not building a Tesla factory or a BYD gigafactory. The engineering goal is to **import proven patterns** into a Kenya-first, M-Pesa-native, modular home-energy line.

2.1 Tesla — system integration and the energy software layer

Tesla pattern	Engineering lesson for Precifarm	Precifarm application
Powerwall — wall-mounted pack + integrated inverter/gateway; stackable capacity over time	Separate **energy** from **power conversion**; customer adds kWh without replacing the inverter	**P2 Power Core** fixed; **Energy Modules** stack under or beside
Megapack — repeatable 3 MWh blocks in a shipping-container form factor	Standard module + repeated mechanical bay = factory and field scalability	**Pod** = N × identical **Energy Modules** in a serviceable rack
Energy segment economics — high-margin software + install network, not commodity cells	Own **Energy OS**, diagnostics, PAYGO and service — not cell chemistry	Energy OS owns runtime prediction, module inventory, remote fault codes
Gateway / Powerwall Backup Switch — single customer-visible "energy brain"	One app account, one telemetry schema across P1 → P2 → Pod	Common device model in Energy OS; SKU-specific capability flags
NACS / open charging network (post-2023)	Open **interfaces** where they create adoption; own **experience** where it creates retention	Module mechanical/electrical interface documented for internal SKUs first; public OEM licensing only if volume justifies
Over-the-air updates	Field-fixable BMS parameters and inverter curves without truck roll for every bug	Signed OTA to Power Core and Pod controller; module firmware pinned and versioned

Reject for Phase 1: proprietary vehicle lock-in, full vertical cell manufacturing, and marketing V2G before Kenya utility rules and bidirectional hardware exist.

2.2 BYD — modularity, $LiFePO_4$ and cost through repetition

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BYD pattern	Engineering lesson for Precifarm	Precifarm application
Blade Battery — long prismatic $LiFePO_4$ cells, cell-to-pack, structural rigidity	**One cell format + one module footprint** repeated across products reduces NRE and spare-parts SKU count	Standard **2.56 kWh Energy Module** built from one prismatic cell family
CTB / modular pack philosophy	Minimize unique pack designs; maximize shared BMS firmware and CAN objects	Single BMS application profile with product-specific limits (max charge C-rate, temp derating)
Vertical integration (~30% cost leverage at OEM scale)	Precifarm **integrates** at enclosure, harness, thermal and software — buys cells and power stages	BOM strategy §8; local assembly partner for Pod racks
DM-i / range-bridge thinking	In weak-grid markets, **right-size storage + solar** before oversizing inverter kW	P2 base: 2.56 kWh / ~3–5 kW inverter class; expand kWh before kW unless load study says otherwise
Export via CKD / local assembly	Pod **rack + Power Unit** suitable for Kenya SKD (cells imported, pack assembled locally when duty/logistics justify)	Phase C: local module assembly line tied to KEBS/UN38.3 batch testing

Reject for Phase 1: claiming Blade-equivalent cell IP, owning prismatic cell R&D, or sizing Pod arrays without per-site load and export-limit studies.

2.3 Synthesis — the Precifarm engineering thesis

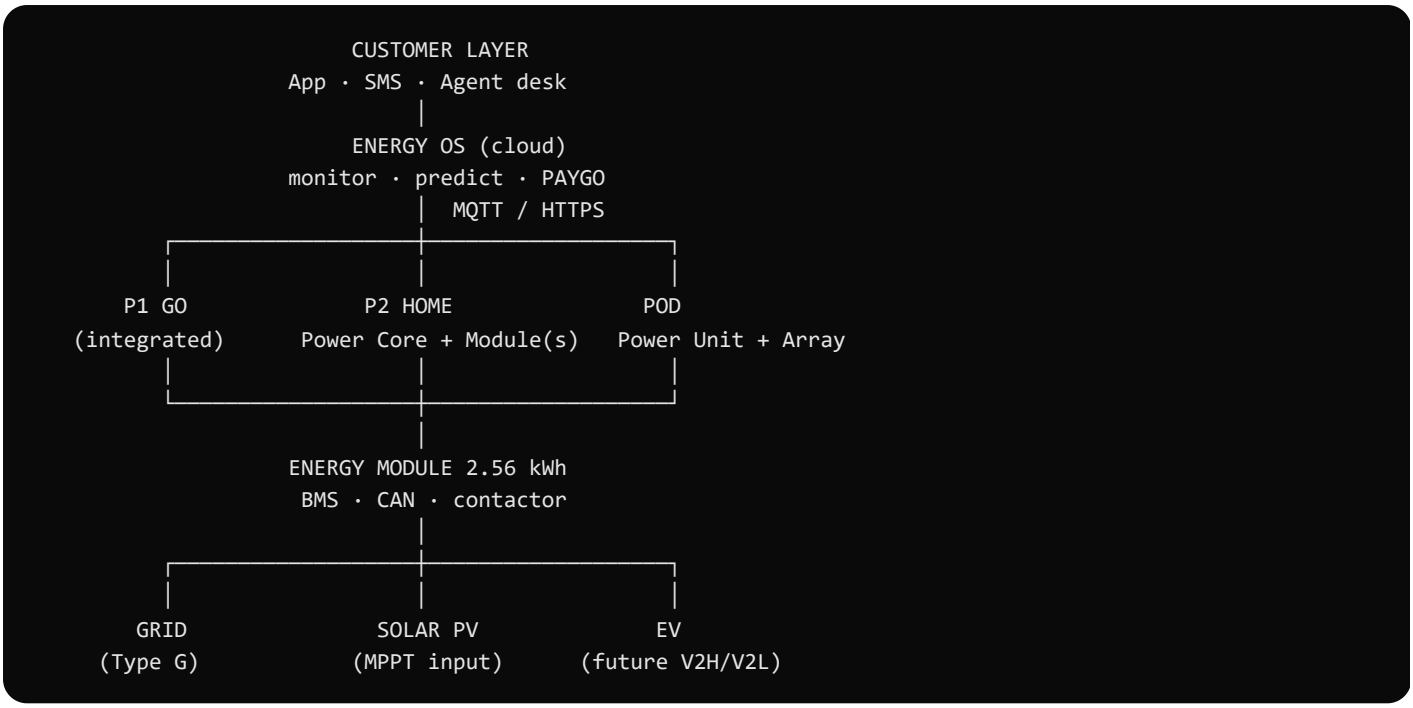
```
TESLA                       BYD                       PRECIFARM
─────                       ───                       ─────────
Integrated experience   +   Repeatable module    =   ONE Energy Module
Software-defined energy +   LiFePO₄ safety       =   Energy OS + PAYGO
Stackable kWh           +   Cost via repetition  =   P2 expand / Pod scale
Open where strategic    +   Local assembly path  =   Kenya SKD when gated
```

North star: Tesla-grade **customer-visible simplicity** and BYD-grade **module repetition** — on African grid reality, M-Pesa rails and Precifarm service network.

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3. System reference architecture

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3.1 Layer definitions

Layer	Responsibility	Failure mode
Energy Module	Store energy; report SOC/SOH; enforce cell limits; disconnect on fault	Module opens contactor; Power Core sees `MODULE_FAULT`
Power Core / Power Unit	Invert/rectify; MPPT; grid sync; load/backup switching; system protection	Trip to safe state; preserve module isolation
Edge controller	CAN aggregation; Wi-Fi/LTE; local rules; OTA agent; tamper detect	Fail-safe: modules idle; passthrough grid if UPS path allows
Energy OS	Fleet telemetry; customer UX; PAYGO; service tickets; analytics	Device continues local safe operation offline

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4. Energy Module — target specification

The **Energy Module** is the shared building block across P2 and Pod (P1 Go uses a smaller integrated pack — see §6).

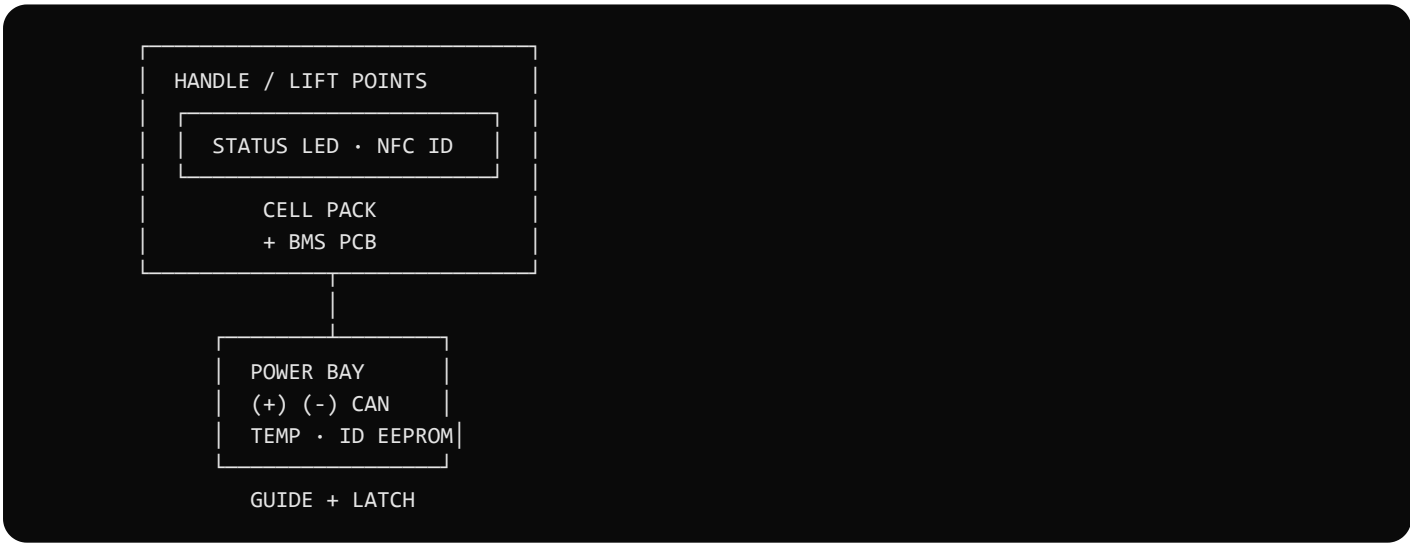
4.1 Electrical targets

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Parameter	Target	Notes
Nominal energy	**2.56 kWh**	Planning basis: 51.2 V × 50 Ah
Chemistry	**$LiFePO_4$**	Prismatic or cylindrical — cell choice is procurement, not architecture
Nominal voltage	**51.2 V**	16s $LiFePO_4$ class
Voltage range	44.8 – 58.4 V	BMS hard limits; firmware derating below/above
Max continuous discharge	**50 A** (~2.5 kW)**	Product may derate P1; Pod aggregates modules
Max continuous charge	**30 A** (~1.5 kW)**	C-rate tuned for cycle life in Kenya heat
Contactor	Dual redundant where cert requires	Precharge on hot-plug
Isolation	≥ 500 V DC isolation module-to-chassis	Hi-pot in production

* Values subject to thermal derating above 45 °C ambient.

4.2 Mechanical and interface



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Interface	Specification
Mechanical	Stackable footprint; guided rail; positive latch; mis-insertion prevention
Electrical	Touch-safe until mated; precharge sequence; keying by module generation
Communication	**CAN 2.0B** 250 kbit/s (baseline); device address from module serial
Identification	Factory-programmed UUID; QR/NFC for service; anti-cloning signature in BMS
Thermal	Module-level NTC grid; passive cooling Phase 1; forced air in Pod rack

4.3 BMS functional requirements

- Cell voltage, temperature, current sampling ≥ 10 Hz for protection; 1 Hz for telemetry
- SOC estimation (coulomb counting + OCV correction at rest)
- SOH estimation (capacity fade model; cycle count)
- Over/under-voltage, over-current, short-circuit, over-temperature, under-temperature
- Contactor weld detect; precharge fail timeout
- CAN heartbeat 1 s; fault latch with manual or service reset policy
- Firmware image **A/B** with rollback; signed updates from Power Core only

4.4 CAN object model (baseline)

PGN / ID range	Content
`0x100–0x10F`	Module status: SOC, SOH, V, I, T_max, fault flags
`0x110–0x11F`	Module limits: max charge/discharge current
`0x120–0x12F`	Module identity: serial, firmware, chemistry code
`0x200–0x20F`	Power Core commands: enable, set current limit, reset fault
`0x300+`	Product-specific (P2 expansion bay enumeration, Pod rack slot)

Document full DBC in repository `docs/product/dbc/` when prototype locks (placeholder — not in v1.0).

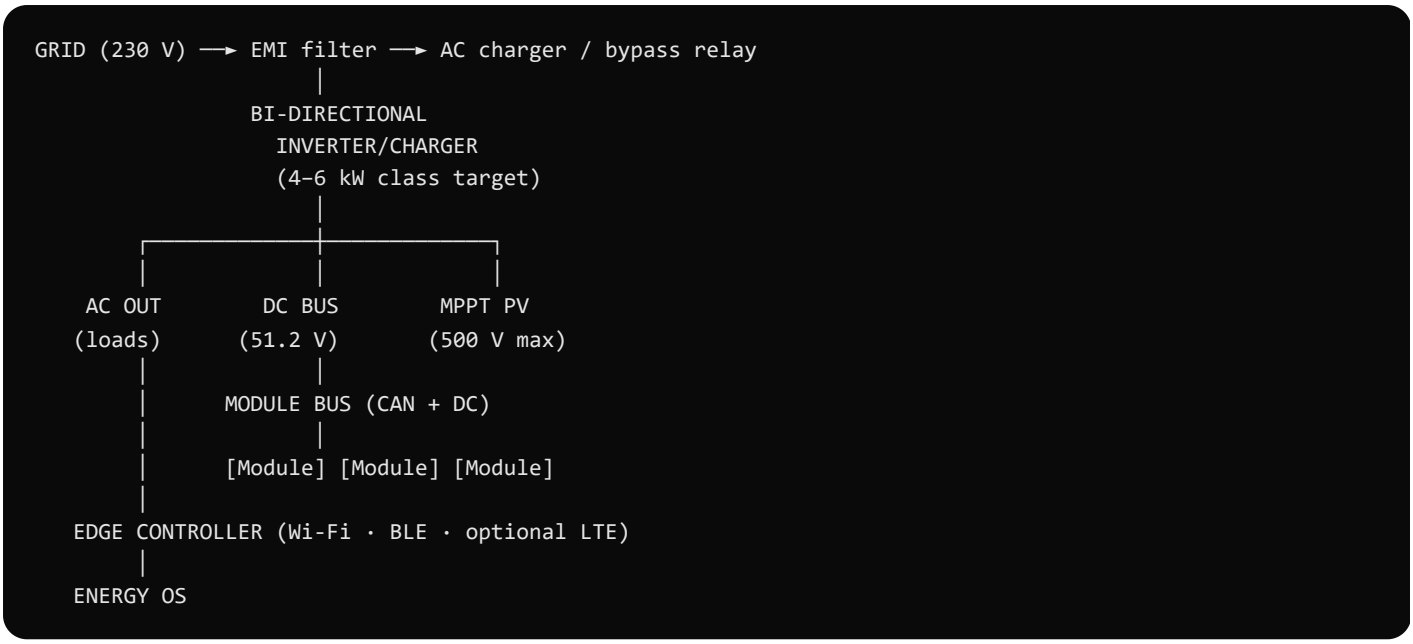
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5. Power Core (P2 Home) — engineering design

P2 separates **conversion and control** from **storage**.

5.1 Block diagram

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5.2 Target ratings (P2 Home base SKU)

Parameter	Phase 1 target	Expansion notes
Inverter continuous	**4 kW**	6 kW SKU via Power Core swap, not module swap
Surge / motor start	**7 kW / 10 s**	Load classification in Energy OS
PV input	**2 kWp** MPPT	1× string; expand input on Pod class
Grid connection	**230 V single-phase**	Kenya domestic; export limit software-configurable
Backup mode	Critical loads panel or whole-home (site-dependent)	Transfer switch external or integrated SKU TBD
Module bays	**1 base + 3 expansion**	2.56 → 10.24 kWh without replacing Power Core
Wheels / enclosure	IP21 indoor; IP54 outdoor variant gated	Thermal derating table per enclosure

5.3 Control logic (edge)

Priority order (local, deterministic):

1. Safety interlocks and emergency stop

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5. Solar self-consumption optimization
6. Grid import limit (customer setting)
7. Cloud advisory schedule (Energy OS)

Cloud **cannot** override safety or PAYGO lockout without signed service token.

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6. P1 Go — integrated portable SKU

P1 is intentionally **integrated** (Tesla Powerwall is wall-integrated; Jackery-class portables are fully integrated). It does **not** use the 2.56 kWh module in Phase 1 — it shares **Energy OS** and UX language only.

Parameter	Target
Energy	**~1 kWh** $LiFePO_4$
Inverter	**600 W–1 kW** pure sine
Ports	230 V AC · USB-C PD · USB-A · 12 V DC
Solar in	**100–200 W** portable panel input
Comms	Wi-Fi + BLE
Mass target	**≤ 12 kg** (carry comfort)
Cert path	IEC 62368-1 / local KEBS alignment for appliance class

Engineering note: P1 firmware still reports to Energy OS using the same telemetry schema with `product_class=P1_GO` and reduced capability flags.

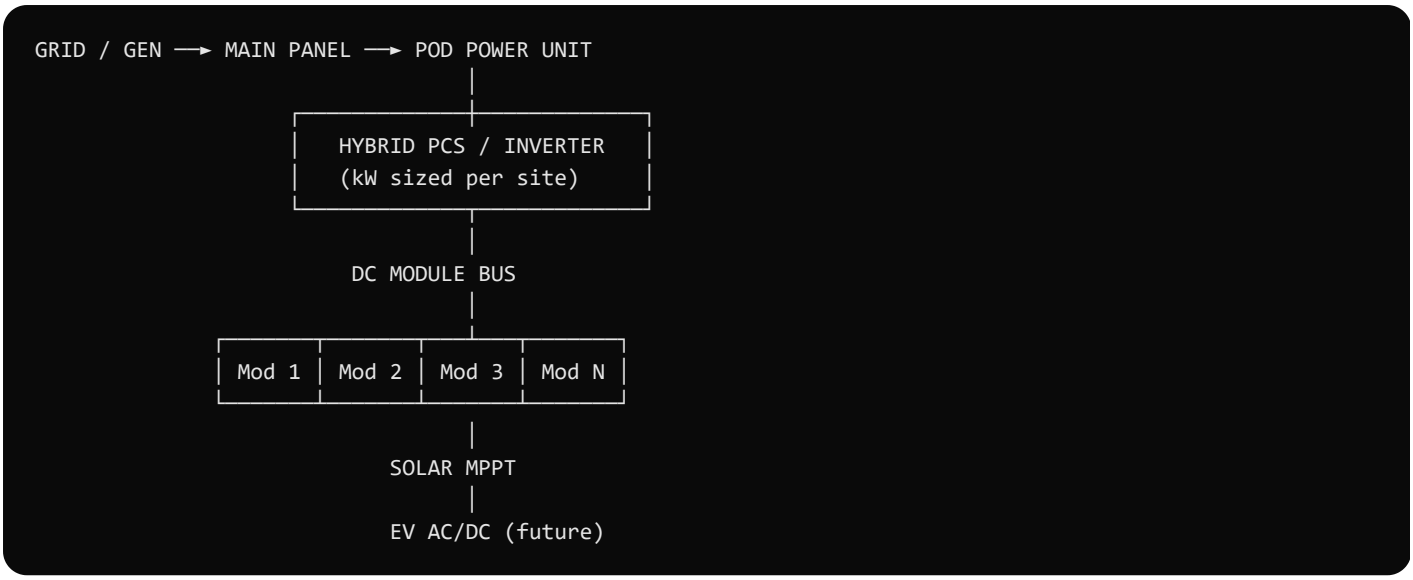
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7. Pod — installed modular infrastructure

Pod applies the BYD **repeat-the-block** pattern at building scale.

7.1 Architecture

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7.2 Sizing method

1. **Load study** — peak kW, daily kWh, backup duration, EV add-on
2. **Energy modules** — round up to N × 2.56 kWh to meet kWh target + reserve
3. **Power unit** — size inverter/PCS to min(peak load, grid export limit, budget)
4. **Solar** — size to daily recharge of target SOC band; not oversized without export policy
5. **Thermal** — rack airflow model; module spacing; Nairobi 25–35 °C roof install derating

Example planning matrix (not a product catalog):

Site class	Modules	Energy	PCS target	Typical use
Home base	2	5.12 kWh	5 kW	Backup + solar
Home premium	4–8	10–20 kWh	8–10 kW	Solar + EV AC
SME	8–20	20–51 kWh	15–30 kW	Shop + cold chain

7.3 Installation envelope (Kenya)

- Licensed electrical contractor; EPRA-aligned workmanship
- Earth electrode and surge protection per site study
- Fire egress and battery compartment ventilation per manufacturer install manual
- Precifarm **survey** → **design** → **install** → **commission** → **expand** workflow mirrored in CMS install pipeline

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8. BOM strategy — buy, customize, own

Aligned with Modular Energy Platform §20.

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Category	Strategy	Examples
Buy	Commodity, multi-source	$LiFePO_4$ cells, contactors, fuses, MPPT ICs, Wi-Fi modules, displays
Customize	Precifarm mechanical + harness	Module enclosure, Pod rack, P2 chassis, thermal paths, labels
Own	Defensibility	Energy OS, CAN schema, PAYGO logic, diagnostics, service procedures

Tesla lesson: integrate the experience. **BYD lesson:** don't integrate the mine — integrate the **repeatable module**.

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9. Energy OS — engineering interface

9.1 Device telemetry (minimum)

Field	Rate	Use
`soc_pct`, `soh_pct`	60 s	Customer UX
`power_w`, `voltage_v`, `current_a`	5 s	Monitor
`runtime_minutes_est`	60 s	Positive UX copy
`fault_code`, `fault_severity`	on event	Service
`module_count`, `module_slots`	on change	Expansion detection
`paygo_state`	on change	Finance gate
`firmware_version`	24 h	OTA compliance

9.2 OTA policy

- Dual-bank firmware on Power Core and Pod controller
- Modules updated only through Power Core proxy (never direct cloud-to-module in Phase 1)
- Staged rollout: canary fleet → region → full
- Rollback if fault rate > threshold within 24 h

9.3 Integration with Precifarm CMS

Install pipeline stages (`survey_requested` → `commissioned`) map to engineering gates:

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CMS stage	Engineering deliverable
Survey requested	Load questionnaire + photos
Survey scheduled	Site visit checklist
Quote sent	Module count + PCS sizing PDF
Deposit paid	Procurement release
Install scheduled	Install pack + SLD
Commissioned	Sign-off test sheet + Energy OS provisioning

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10. Safety, standards and certification path

10.1 Target standard families

Domain	Reference standards	Application
Battery safety	IEC 62619, UN 38.3	Module transport and stationary
BMS	IEC 61508 concepts (SIL goals TBD)	Fault handling
Inverter/grid	IEC 62040 / local grid code	Backup and export
EMC	IEC 61000 series	Residential
Appliance	IEC 62368-1	P1 portable class

Kenya **KEBS** conformity and **EPRA** electrical installation rules apply at deployment — engage local cert body before mass retail.

10.2 Failure modes and effects (summary)

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Failure	Detection	Response
Cell over-temp	BMS NTC	Derate → open contactor
Contactor weld	Voltage mismatch	Alarm; disable charge/discharge; service call
Ground fault	RCD / insulation monitor	Trip; log to Energy OS
CAN loss	Heartbeat timeout	Module idle; redundant polling
Grid loss	Voltage sag	Backup transition if configured
PAYGO default	Energy OS	Grace period → export limit / discharge limit policy (legal review)

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11. Thermal and environmental design

Kenya install ambient **15–40 °C** operating; design for **45 °C** derated performance.

Product	Cooling strategy
P1 Go	Passive fins; no fan if possible (dust/noise)
P2 Home	Passive base + ducted fan on inverter compartment
Energy Module	Passive in open bay; forced airflow in Pod rack
Pod outdoor	IP-rated enclosure; sun shield; minimum module spacing per CFD

BYD lesson: large thermal mass and conservative C-rates beat chasing peak kW in hot climates.

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12. EV integration (future interface)

Precifarm does not block V2X; it **sequences** it.

Phase	Capability	Hardware
A	AC EV charging from Pod/P2 surplus	Type 2 / Pulse charger integration
B	Monitoring EV SOC via OEM API where available	Energy OS only
C	V2H / V2L	Bidirectional charger + vehicle compatibility list
D	V2G	Utility programme + EPRA approval

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13. Prototype and validation roadmap

Phase	Duration	Deliverable	Gate
P0 — Module breadboard	8–12 weeks	Single 2.56 kWh module + BMS + CAN dev harness	500 cycle lab test plan started
P1 — P2 alpha	12–16 weeks	Power Core + 1 module; backup + solar MPPT	Internal dogfood 30 days
P2 — Pod pilot	16–24 weeks	4-module rack; 2 home installs Nairobi	Commissioning checklist pass
P3 — Cert + pilot retail	6–9 months	KEBS path; 20 P2 + 5 Pod paid installs	Fault rate < target; PAYGO working
P4 — SKD evaluation	gated	Local module assembly business case	Duty/logistics vs import

14. Open engineering decisions

ID	Decision	Options	Target date
E-01	Cell form factor	Prismatic vs cylindrical	P0 exit
E-02	P1 integration vs mini-module	Integrated only vs 0.64 kWh sub-module	P1 kickoff
E-03	Grid export policy default	Zero-export vs limited export	Legal + EPRA
E-04	PAYGO discharge limit ethics	Hard cutoff vs essential-load only	Product + legal
E-05	LTE vs Wi-Fi-only P2	SKU split	After Nairobi connectivity study
E-06	Pod PCS supplier	Hybrid inverter OEM shortlist	P2 parallel track

15. Document control

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Version	Date	Author	Change
1.0	2026-08-30	Precifarm Engineering	Initial engineering design basis; Tesla/BYD motivation; module spec
1.0-s	2026-08-30	Precifarm Engineering	Marked superseded by PF-MODENERGY-002 v2.0 — do not update targets here

Related documents

- **Modular Energy Architecture v2** — PF-MODENERGY-002 (current)
- Modular Energy Platform — Product Design
- Solar Chargers & Stations Engineering
- Software & Cybersecurity Architecture
- Canon

Precifarm Modular Energy — engineering targets for prototype and certification planning. Not a product warranty or customer specification.

Precifarm
Electric mobility infrastructure for Africa

Modular Energy Storage — Product & Engineering Design

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sales@precifarm.com · +254 794 702 768 · Nairobi, Kenya · precifarm.com

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